QUESTION 1 [15 marks]

a. **(9 marks)** For the circuit shown in Figure 1,

i. **(6 marks)** Calculate the equivalent resistance R_{eq} as seen from terminal *a*-*b*.

ii. **(3 marks)** Find the voltage drop V_0 using the result of part **(i)**.

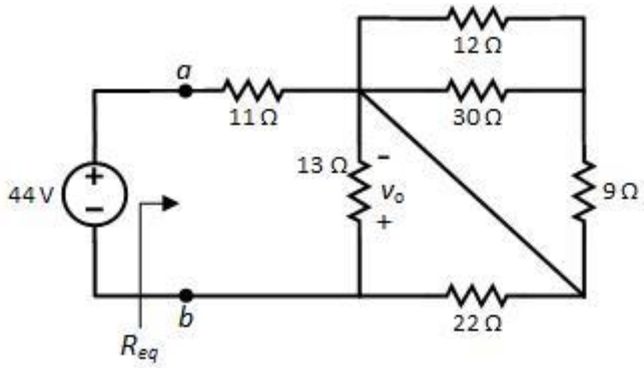

Figure 1

b. **(6 marks)** A kitchen has three appliances connected to a 120 V circuit with a 15 A circuit breaker: an 800 W coffee maker, a 1200 W microwave and an 850W toaster, as shown in Figure 2. Which of these appliances can be operated simultaneously without tripping the circuit breaker? Note that a circuit breaker trips when the circuit draws more current than its amperage rating.

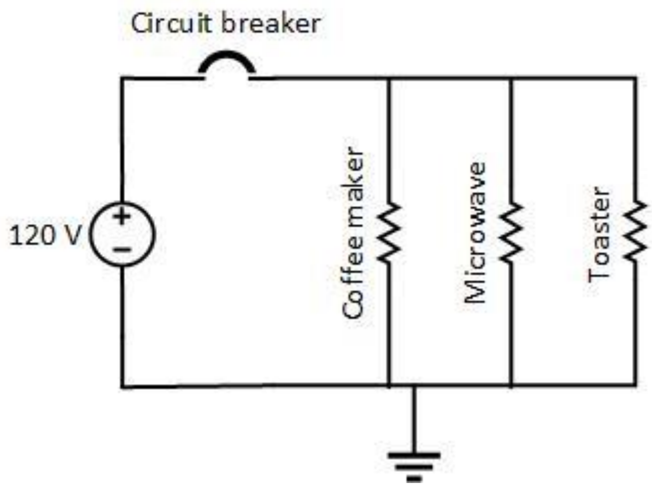

Figure 2

QUESTION 2 [20 marks]

The circuit in Figure 3 represents a solar-powered water fountain. The water fountain is powered by a solar panel during the day, and by a battery at night. The wiring resistances are as shown in Figure 3.

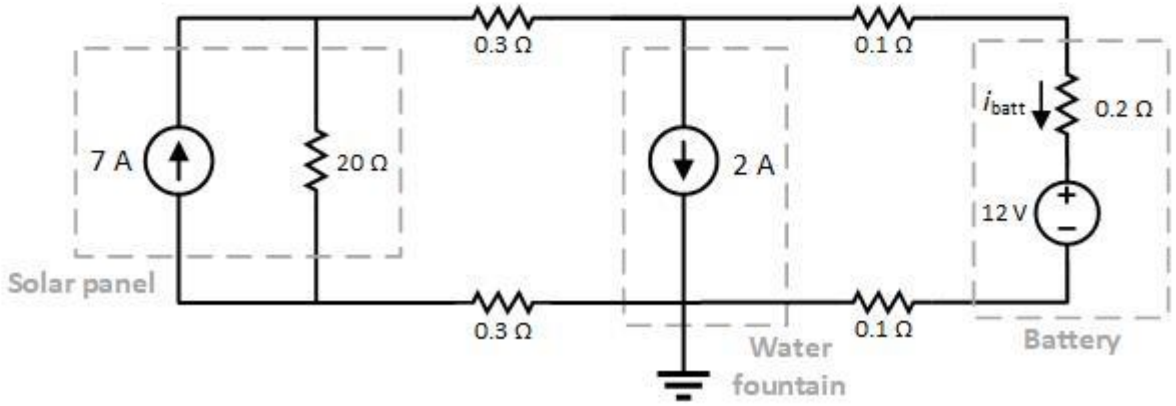

Figure 3

a. **(10 marks)** Calculate the current through the battery, i_{batt}, during the daytime using mesh analysis.

b. **(5 marks)** How much total energy will be available from the battery if, during the day, the solar panel delivers its rated current (7A) for 8 hours?

c. **(5 marks)** Calculate the power of the solar panel.

QUESTION 3 [25 marks]

a. (12 marks) For the circuit shown in Figure 4, use superposition to calculate the voltage v_x.

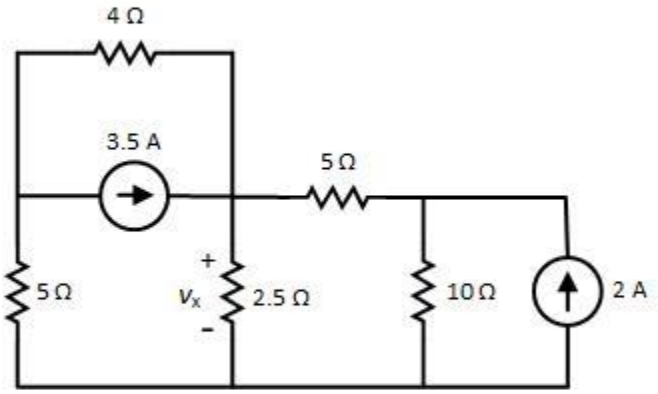

Figure 4

b. (13 marks) Apply nodal analysis and write down the nodal equations using the labelled node voltages.
<u>Note</u>: Simplify the equations in the form $aV_1+bV_2+cV_3=d$, where coefficients a, b, c and d can be positive or negative numbers, but **DO NOT** solve them.

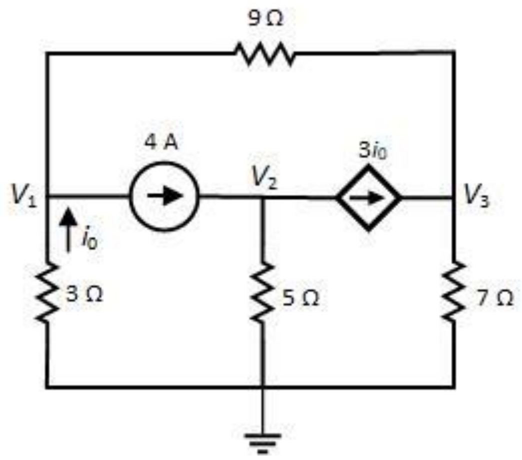

Figure 5

QUESTION 4 [20 marks]

The circuit shown in Figure 6 is being used to power a Christmas lighting system. The lighting system can be modelled as a single resistor connected to terminals *a*-*b.*

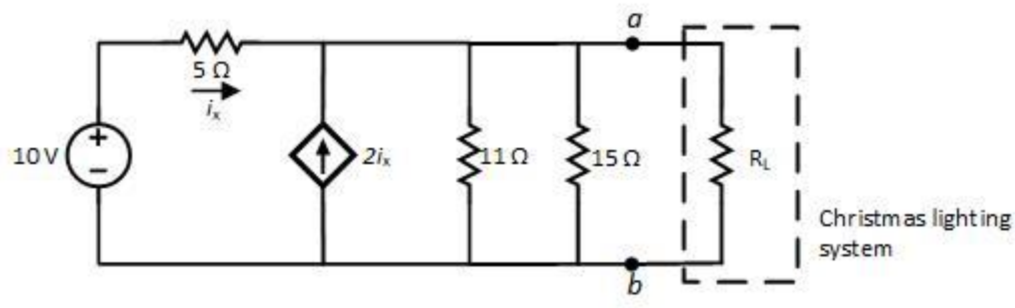

Figure 6

a. (9 marks) Calculate the Thevenin voltage at terminals a-b.

b. (9 marks) Calculate the lighting system resistance that will ensure the maximum transfer of power from the circuit to the lights.

c. (2 marks) Find the maximum power that can be delivered to the lights.

QUESTION 5 [20 marks]

a. **(8 marks)** Find the currents i_1 and i_2 and voltage v_o in the circuit of Figure 7 under DC steady-state conditions.

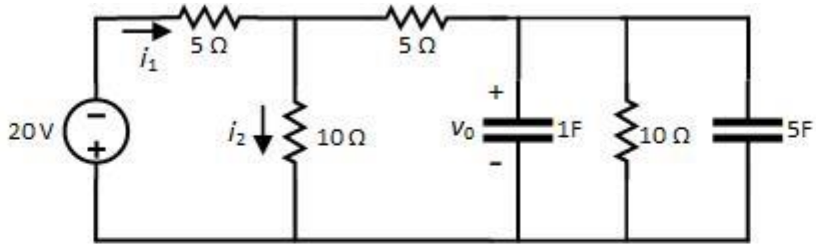

Figure 7

b. **(12 marks)** In the circuit below, the switch has been open for a long time and it is closed at t=0.

i. **(10 marks)** Find v(t) for t > 0.

ii. **(2 marks)** Calculate v(t) at t=0.2s.

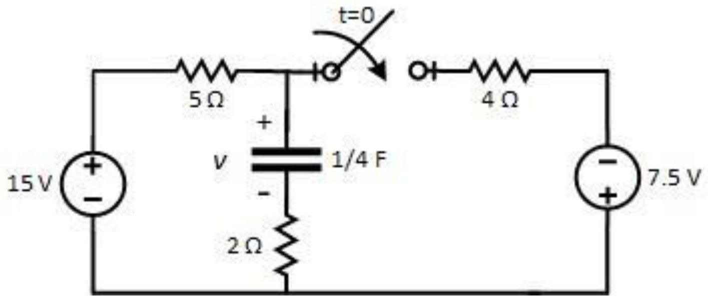

Figure 8